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PI Name:	Dynlacht, Joseph Robert Ph.D.		
Project Title:	Effects of Estrogen on Cataract Induction After Exposure to High LET Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
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Space Biology Special Category:	None		
PI Email:	jdynlach@iupui.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	(317) 278-3882
Organization Name:	Indiana University		
PI Address 1:	Radiation Oncology, RT-041		
PI Address 2:	Indiana Cancer Pavilion, 535 Barnhill Drive		
PI Web Page:			
City:	Indianapolis	**State:**	IN
Zip Code:	46202-5121	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
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Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: PI to receive NCE to 8/31/2010 (from 8/31/2009) per J. Dardano/JSC (7/09)		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Mendonca, Marc (Indiana U) Valluri, Shelia (Indiana U) Bigsby, Robert (Indiana U)		
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Task Description:	Several studies have demonstrated that there is an increased cataratogenic potential in animals and humans exposed to high linear energy transfer (LET) radiation compared to low LET radiation. Since the relative biological effectiveness (RBE) of heavy energetic charged (HZE) particles for lens opacification is high, these studies imply a high cataract risk for astronauts involved in long-duration space missions. During a manned mission to Mars, ocular doses in excess of ~0.1 Gy of HZE particles could be accumulated. Though not directly life-threatening, radiation-induced cataractogenesis represents a potentially serious sequelae which could necessitate surgical intervention in the years after, or perhaps even during the mission in some individuals (see below). While the cellular and molecular mechanism(s) of radiation-induced cataractogenesis have not been clearly elucidated, damage to DNA at the time of exposure and subsequent proliferation of the radiosensitive cells in the germinative zone of the lens epithelium likely play a role in the process. Using a rat model, we have recently accumulated data which indicate that estrogen increases or reduces the latent period and may modulate the incidence and severity of cataracts induced by low LET radiation, depending upon when the estrogen is administered. High estrogen levels are induced in non-pregnant women using oral contraceptives, or in post-menopausal women on estrogen replacement therapy. Therefore, sub-groups of female astronauts may be at an increased risk for developing cataracts that are more severe or occur with a more rapid onset. The interaction between estrogen and high LET radiation has never been studied before in the context of cataractogenesis. Therefore, a mechanistic approach characterizing the modulating effect of estrogen on high LET radiation-induced cataractogenesis is proposed. The underlying hypothesis of this proposal is that estrogen alters the latent period for high LET radiation-induced cataractogenesis, and, depending upon when the it is administered, modulates severity and incidence of cataracts. Estrogens regulate several proteins involved in cell cycle control and apoptosis, and it's metabolism results in the production of free radicals which may be genotoxic and mutagenic to mammalian cells. Thus, a novel hypothesis to be tested in the proposed studies is that estrogen alters cell cycle regulation, DNA double strand break (DSB) induction or repair, and proliferation in lens cells exposed to HZE particles (1 GeV/amu iron ion beams). The first Specific Aim shall be to define the dose-time interactions of high LET radiation (vs. low LET radiation) and estradiol to better understand the mechanism of estrogen action. The RBE for cataractogenesis in estrogen-treated animals exposed to1 GeV iron beams shall be determined using 60-Co gamma rays as a standard. The second specific aim shall be to determine the mechanism of estrogen-modulation of radiation-induced cataracts. Animal and cell culture studies will be performed to determine the effects of estrogen on gene expression, cell proliferation, apoptosis, and the induction and repair of radiation-induced DSBs. Slit-lamp biomicroscopy will be used to measure lens opacification in exposed animals. Successful completion of these studies will lead to a greater understanding of the modulation of space radiation effects by hormones and novel intervention strategies for long-duration space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results obtained from this project may provide greater insight into the potential effects of hormones on normal tissues after radiotherapy treatments with high LET particles.
Task Progress:	This report summarizes progress during the third full year of support. As in the previous two years, work was focused on the first Specific Aim of the project, which is to determine whether estrogen (17-beta-estradiol, E2) alters cataractogenesis induced by high LET radiation. During the third year of funding, we completed our third and fourth "runs" at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL); beam time had been requested in the Fall 2007 and Spring 2008 runs. We have continued to use our specially designed animal holder and collimator set-up to irradiate the right eyes of up to 3 rats simultaneously, while sparing the contralateral eye and most other normal tissues. This resulted in other structures receiving a dose of less than 5% of the dose to the target eye. Over three hundred rats have been irradiated in this manner, and we have yet to observe a radiation-induced cataract in the contralateral eye. We completed our accrual of data from an experiment that was initiated during Year 1, prior to our first runs at NSRL. Male and female rats were exposed to 60-Co gamma-rays in order for us to assess gender-based differences in cataractogenesis after exposure to sparsely ionizing (or low LET) radiation. Placebo-implanted or E2-implanted 56-day old male rats and ovary-intact placebo-implanted 56-day old female rats were irradiated at Indiana University with 10 Gy using the Leksell Gamma Knife to irradiate only the right eye of each test animal with 60-Co gamma-rays. Slit-lamp biomicroscopy was used to measure lens opacification in exposed animals. Opacification of the anterior and posterior subcapsular (ASC and PSC, respectively) lenses of irradiated and control (contralateral) eyes was measured every 2-4 weeks. In addition to permitting an assessment of gender differences for cataractogenesis, the experiment also allowed us to determine whether estrogen would modulate radiation cataractogenesis. We found no difference in the severity and progression of radiation-induced cataracts between male placebo- or E2-implanted rats, and female rats. However, there was a significant difference in the incidence of anterior and posterior subcapsular cataracts between male and female placebo-implanted rats. Also, male rats with placebo implants showed no difference in cataractogenic potential when compared to male rats with E2 implants which were exposed to low LET radiation. Thus, a gender effect for radiation cataractogenesis was noted, but this was not due to estrogen. During the Fall 2006 run at NSRL, we initiated an experiment similar to those described above that will allow us to determine whether there is a similar gender effect after exposure to high LET radiation. Data accrual for this experiment is nearly complete. Specifically, we tested the hypotheses that potential gender-based differences in incidence and latent period for, or progression of cataractogenesis after exposure of rats to high LET radiation are related to steroid hormones. Placebo-implanted or estrogen-implanted 56-day old male rats and ovary-intact placebo-implanted 56-day old female rats were irradiated with 1 Gy of a 600 MeV/amu 56-Fe beam during the Fall 2006 run. Appropriately treated sham-irradiated control rats were also sent to the NSRL facility. The right eyes of ovary-intact female rats or male rats implanted with either empty capsules or capsules containing E2 were irradiated with 1 Gy of 600 MeV 56Fe ions. Rates of PSC progression were found to be significantly greater in intact female rats and male rats with E2 when compared to males rats with empty capsules implanted. ASC progression showed the same trend as PSC progression. Interestingly, the incidence of cataracts (either ASC or PSC) was significantly increased in intact female rats and male rats with E2, compared to male rats with empty capsules implanted. Thus, the gender difference was reversed for rats exposed to high LET radiation (1 Gy of 600 MeV 56Fe ions), with female rats showing enhanced cataractogenesis compared to male rats. This enhancement was due to estrogen.

Beginning during the Fall 2006 run and continuing through the Fall 2007 run, a radiation dose response experiment was initiated at NSRL to determine whether modulation of cataractogenesis by E2 after exposure to high LET radiation is dependent upon the radiation dose. Ovariectomized or intact female rats, or ovariectomized rats that received estrogen-implants were irradiated with 0.05 to 2 Gy of 600 MeV/amu 56Fe ions. All doses have been delivered, and although data accrual is almost complete for the 1 and 2 Gy groups (since latent period is inversely related to dose), data accrual is continuing for animals irradiated with lower doses.

However, results with 1 and 2 Gy doses have yielded interesting results. The rate of PSC progression after irradiation with 1 Gy of iron ions was significantly greater in intact females and ovariectomized females with E2, compared to ovariectomized females with empty capsules implanted (ASC & PSC progression was similar). Also, the incidence of cataracts (either ASC or PSC) was significantly higher in intact females and ovariectomized females with E2, compared to ovariectomized females with empty capsules implanted. Thus, it appears that endogenous or exogenous estrogen may enhance both the rate and incidence of cataractogenesis after exposure to high LET radiation. After irradiation with 2 Gy of iron, the rate of PSC progression was significantly greater in ovariectomized females implanted with E2 compared to ovariectomized or intact animals (ASC & PSC progression was again similar). Ovariectomized females with implanted E2 also had a significantly higher incidence of cataract (either ASC or PSC) than the other groups. We conclude that enhancement of radiation cataractogenesis by estrogen appears be dependent upon LET as well as dose.

Additional experiments were performed during the Fall 2007 and Spring 2008 runs to determine the effects of age on radiation cataractogenesis induced by high LET radiation, as well as the effects of timing of the estrogen response. In this latter experiment, the goal is to determine whether estrogen, if implanted after irradiation, can protect against high LET radiation-induced cataractogenesis, as we have observed after exposure to low LET radiation.

When data accrual for all of these experiments is complete, the issue of gender differences in radiation cataractogenesis will be resolved, and may have important implications in regard to age and gender consideration for extended manned space flight.

Bibliography Type:	Description: (Last Updated: 07/17/2025)
Abstracts for Journals and Proceedings	Henderson MA, Valluri S, Rusek A, DesRosiers C, Lopez JT, Batuello CN, Caperell-Grant A, Mendonca MS, Pearson C, Powers E-M, Bigsby RM, Dynlacht JR. "Gender-related Differences in Radiation Cataractogenesis." Presented at the 18th Annual NASA Space Radiation Investigators' Workshop, Sonoma, CA July 2007. 18th Annual NASA Space Radiation Investigators' Workshop, Sonoma, CA July 2007. , Jul-2007
Abstracts for Journals and Proceedings	Henderson MA, Valluri S, Bigsby RM, Mendonca MS, Lopez JT, Caperell-Grant A, Rusek A, Garrett J, Batuello CN, Dynlacht JR. "Estrogen Modulates Cataractogenesis Induced by High LET Radiation." Presented at the 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 2008. 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 2008. , Jun-2008
Abstracts for Journals and Proceedings	Henderson MA, Valluri S, Rusek A, DesRosiers C, Lopez JT, Batuello CN, Caperell-Grant A, Mendonca MS, Pearson C, Powers E-M, Bigsby RM, Dynlacht JR. "Gender-related Differences in Radiation Cataractogenesis." 13th International Congress of Radiation Research, San Francisco, CA, July 2007. 13th International Congress of Radiation Research, San Francisco, CA, July 2007. , Jul-2007